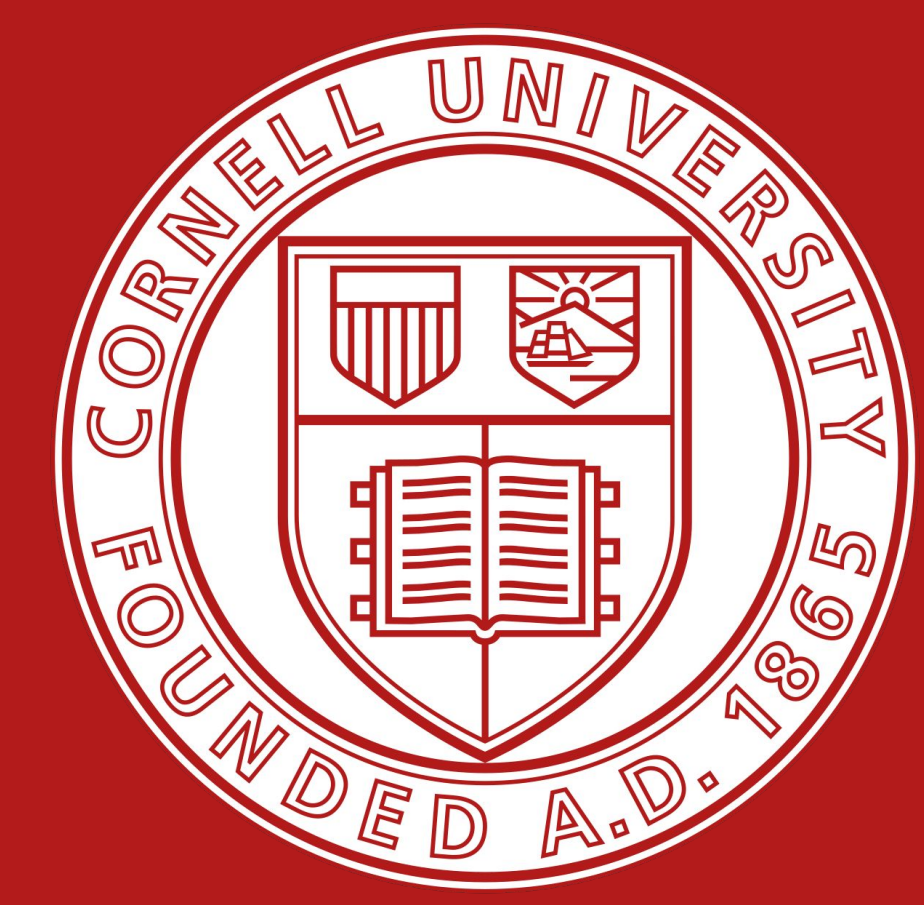




Vibrating Filtration System for Spotted Lanternfly Mitigation in Grape Harvesting

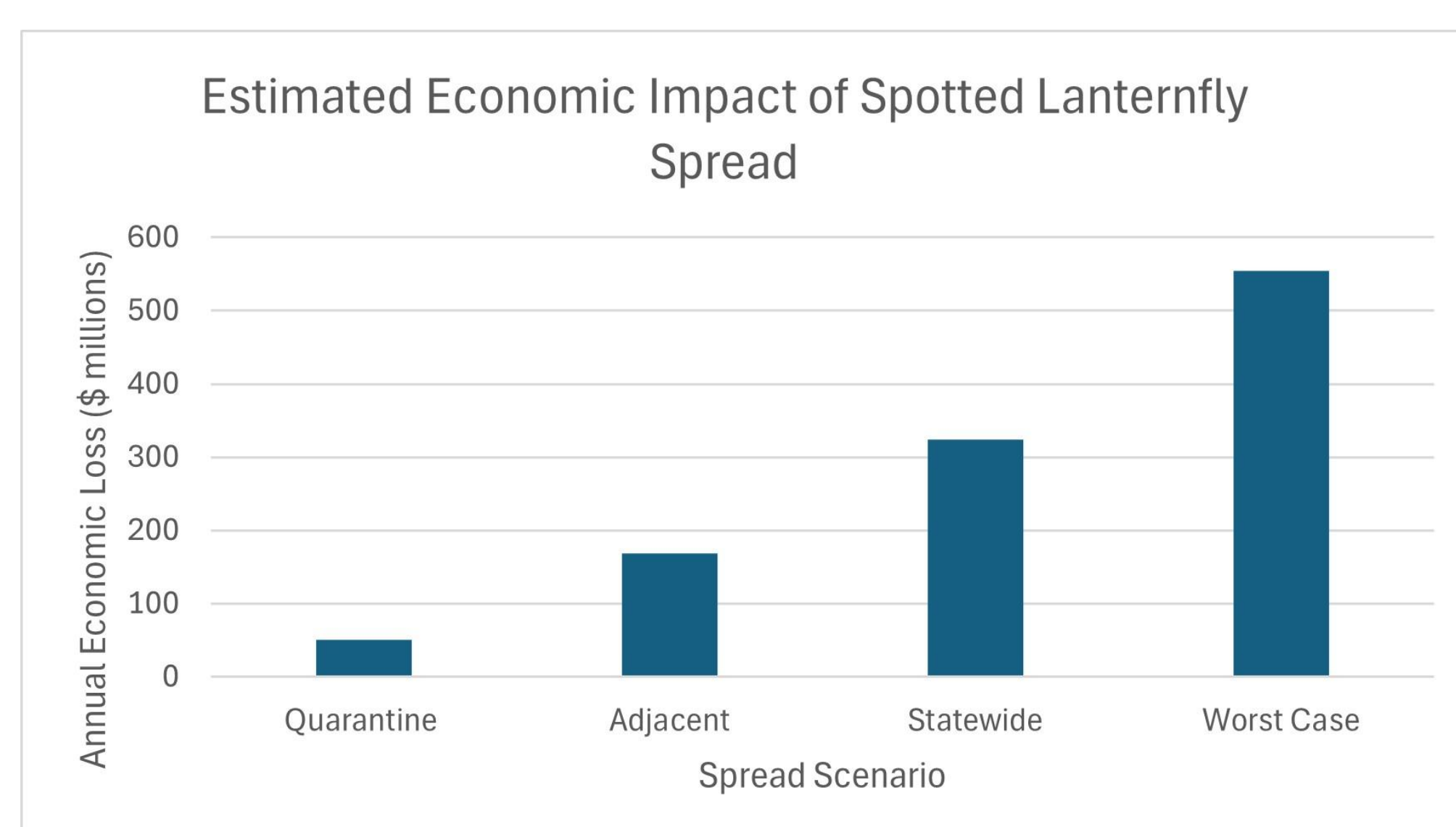
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Introduction & Problem Overview

Spotted Lanternflies (SLF) contaminate grapes during harvesting by **releasing chemicals that taint juice and reduces fruit quality**.

Contaminated grapes often become unsuitable for processing and must be discarded.

Because grapes are sold by weight, **every ounce of usable product directly contributes to revenue**. Any contamination that reduces usable yield results in a measurable financial loss. SLF infestations have already cost major economic damage to agricultural regions. As the infestation spreads, losses increase significantly due to reduced crop quality and lower marketable yield.



Estimated annual economic impact of spotted lanternfly infestation in Pennsylvania (Duke).

Sub-Problem Identified

Spotted Lanternflies presence in collected grapes **during** the harvesting process. Contamination increases with the time spent of the flies in grape mixture.

Design Goal

This design focuses on **saving as much of the usable grape harvest as possible**, even if that means losing small percents of the solid harvest:

- Straining the liquid out allows harvesters to keep the more valuable product
- Separation of solids will allow for simple recognition of SLF in harvest
- Easy removal of SLF contaminated grape harvest from viable product

The design is meant to be simple and adjustable for existing harvester gondolas. It prioritizes a mechanical approach to ensure reliability, easy use and low costs.

The Prototype

The device is a vibrating filtration system designed to sit inside a vineyard gondola during harvesting. Juice passes through while solids are retained. The final prototype consists of:

- **Perforated PVC Tray:** allows juice to pass while retaining solids and SLF material.
- **Adjustable Aluminum Frame:** Adjustable height with pin-lock mechanism
- **Cable Suspension System:** Provides tension support and reduces structural stress.
- **Motor-Driven Vibration Linkage:** Generates controlled vibrations to prevent clogging and improve flow.



Harvester and Field Gondola



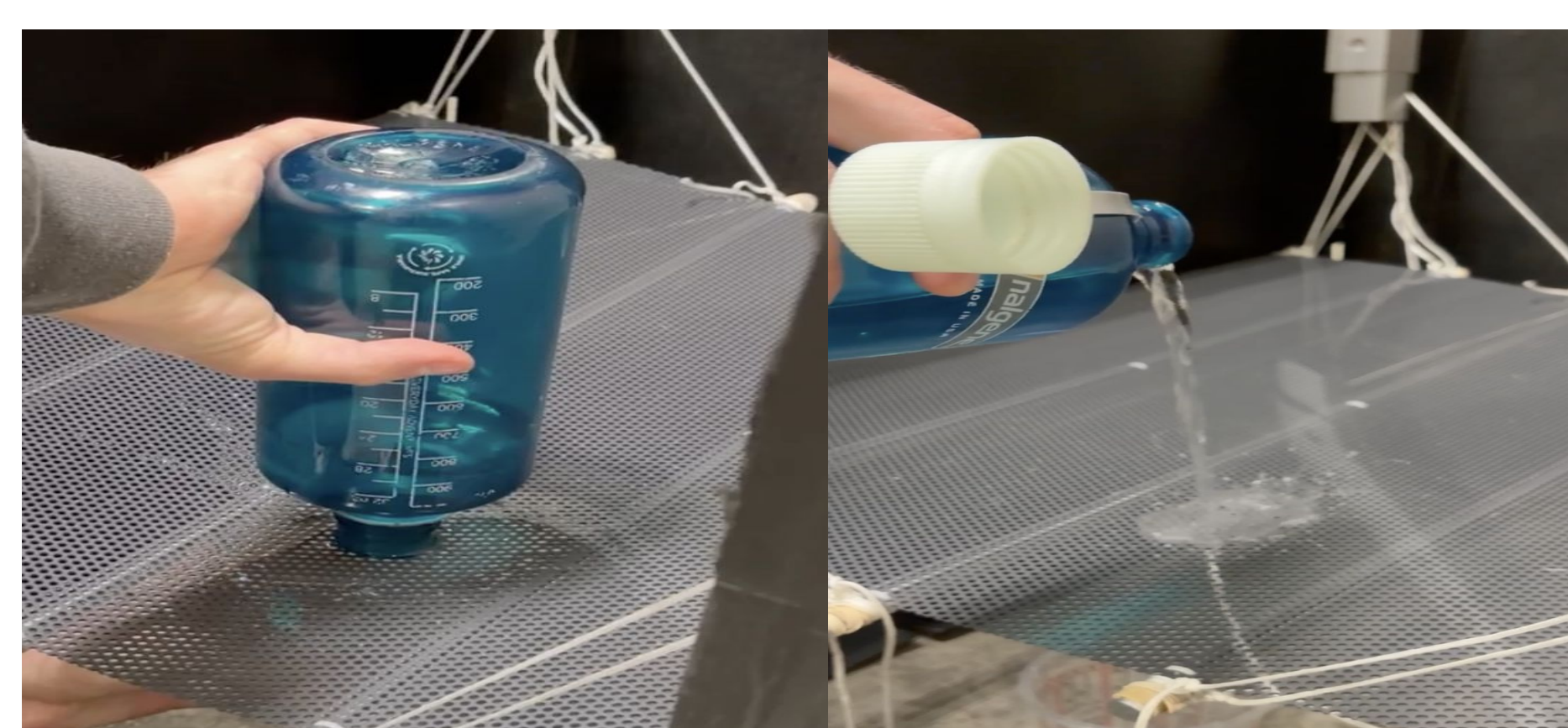
Left: Prototype mounted in mock gondola; Right: view of tray

Testing Methods

Draining Flow Rate:

Flow rate was conducted under two different input conditions to evaluate how input height and flow distribution affect the drainage performance.

- First test: a water bottle was held directly above the sheet keeping the surface flat and water flow vertical (**3.26 ounces per second per square inch**)
- Second test: water was poured several inches above the sheet allowing the spread of flow across the surface.



Left: view of first test; Right: view of second test

Testing Methods (cont.)

Static Load Capacity:

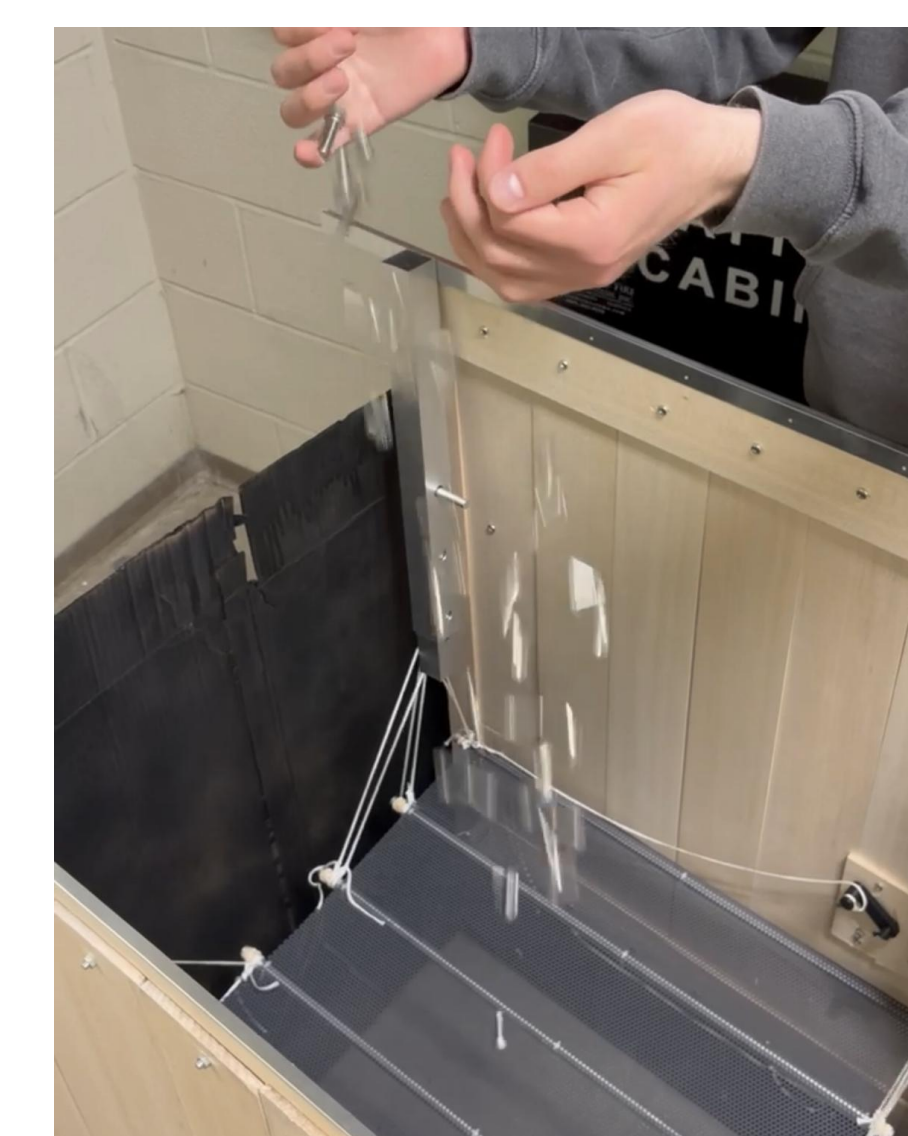
The maximum weight tested was **100lbs**.



View of capacity test at maximum weight

Dynamic Load:

Dynamic load test was performed by releasing bolts from a height of 1m onto the straining surface to simulate falling grapes. Each bolt weighed **14.3g**, with **40 bolts** dropped each trial. The bolts provided an estimate of the impact loading with grapes being approximately 5g each.



View of Dynamic Load Test

Oscillation Amplitude and Frequency:

Oscillation amplitude determined to be **2cm** with frequency of **1.11 Hz**

Conclusion and Future

The final prototype successfully **reduces contamination through separation** and **preserves usable grape juice during harvest**. Testing shows the system is structurally sound, stable, and effective at filtering juice while retaining solids. The following work would be ideal to further develop the prototype:

- **Adjustable Motor Speed Control:** Add speed control to vary vibration frequency and optimize performance.
- **Chemical Validation:** Test collected juice to confirm contamination reduction.
- **Long-Term Field Testing:** Conduct extended trials across multiple vineyards and conditions to evaluate durability and performance.

References

- Ambrose, Kevin, and Kasha Patel. "Invasive Spotted Lantern Fly Spreads Across Mid-Atlantic." The Washington Post, 17 Sep. 2021, p. B2.
- Duke, Amy. "Scientists Examine Potential Economic Impact of Spotted Lanternfly in Pa." Penn State News, 15 Jan. 2020